ATU VIRTUAL LABORATORY

Wheel-Slip Control Using ABS and TCS

A practical wheel-corner laboratory: grip, braking, traction and feedback

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A practical guide from first use to test design. The system is a browser-based educational simulation; physical brake-system and vehicle validation are separate activities.

First experiment

Load ABS emergency braking and press Run simulation. The computer calculates the entire run, then you can press Play or drag Playback position. Read the vehicle and tyre surface speeds together. Compare applied brake torque, slip and the highlighted controller stage at the same instant.

Try Braking without ABS next. Keep the same road, initial speed and torque so it is a fair comparison. For acceleration, compare TCS low-friction launch with Launch without TCS. Export JSON or CSV to keep the complete inputs and 1 ms samples. Editing an input clears the old result; run again before interpreting the display.

Step time in simple terms

Brake step time is the second when the brake request changes. If it is 1 s, brake before is 0 and brake after is 1, the model waits one simulated second and then requests full braking. Throttle step time works the same way for the accelerator.

The integration time step is different: it is fixed at 0.001 s, or one thousandth of a second. The computer updates the equations 1,000 times per simulated second. A 12 s run contains 12,001 saved points including the starting point. This reference step is fixed because changing it changes the numerical experiment.

Playback speed only changes how fast results replay on screen. Drag the scrubber to inspect a precise sample; pause to read the signal loop. Chart zoom enlarges the time window around the chosen playback position. None of these viewing controls alters a calculated result.

What the wheel-corner view shows

The view removes the body and other vehicle systems to focus on the tyre, ventilated brake disc, fixed caliper, rotating encoder ring, wheel-speed sensor, ABS valve block and return-pump motor. It is a representative quarter-corner teaching assembly using the Genesis GV70 as the vehicle reference, not an OEM CAD drawing or a claim about exact GV70 hydraulic routing.

Rotate the scene by dragging. Use Wheel and brake or ABS modulator for a close-up. Tyre, disc and encoder rotation follow the recorded wheel angle. Road markers follow recorded distance. Caliper emphasis follows applied brake torque; the contact patch indicates whether braking or driving slip exceeds the upper intervention threshold. The amber ground arrow shows the signed per-wheel force, scaled to the entered road-friction and normal-load limit. These are explanatory visual encodings, not measured temperature, pressure or contact-patch deformation.

Blue represents ideal sensor feedback; amber represents brake actuation; green represents the drive-torque path. The numerical signal diagram below the 3D view remains available if your browser cannot show 3D. Reduced-motion freezes mechanical movement while keeping all selected-sample values readable.

Follow one feedback cycle

The real-world concept is wheel rotation to wheel-speed sensor, then controller, then actuator, then changed tyre behaviour, then a new sensor reading. The teaching model supplies ideal wheel speed and ideal vehicle speed; it does not simulate electrical sensor pulses, an estimator, CAN messages, sensor noise or communication delays.

For a saved sample after time zero, the controller used speeds and slip from the preceding 1 ms sample. The signal panel shows that earlier sensor time separately from the resulting state time. This avoids implying that the new state caused a command already issued in that same step. At time zero, the displayed state is initial and no controller update has run yet.

The controller compares braking slip with 0.12 and 0.20. Below 0.12 it applies brake torque, between the thresholds it holds, and above 0.20 it releases. ABS is active only with a brake request above 0.001 and vehicle speed above 1 m/s. Repeated updates can move rapidly between these stages.

Return pump and TCS: read the distinction

A real ABS hydraulic circuit can isolate and release brake fluid and use a return pump. This solver changes torque directly; it contains no fluid-pressure, valve-current, pump-speed or reservoir-volume equations. The animation therefore highlights the return path and moves the pump display only during ABS RELEASE as a labelled teaching cue. It must not be read as a predicted pump duty cycle, hydraulic pressure trace or actual GV70 wiring sequence.

ABS APPLY and HOLD remain visible even when the return-pump teaching cue is off. The ABS active flag includes all three modes; it cannot be treated as a pump-on measurement.

For TCS, slip above 0.18 reduces drive torque, slip below 0.10 restores it, and the middle region holds it. This particular model uses drive-torque control only. The drive command path is highlighted; the hydraulic return-pump cue stays off. Brake demand above 0.01 overrides the accelerator request.

How grip and the equations relate

Matched tyre and road speeds mean zero slip. Braking can make the tyre turn too slowly; acceleration can make it turn too fast. Slip is normalised by the larger of vehicle speed, tyre surface speed and 0.5 m/s. The small-speed floor avoids division by zero; it also limits what near-standstill slip means.

The model uses equal normal load m times 9.81 divided by 4. A supplied nonlinear slip curve scales the per-wheel tyre force by the selected road friction. Resistance combines rolling resistance and aerodynamic drag. Vehicle acceleration uses four tyre forces minus resistance, divided by vehicle mass.

The written wheel equation is (drive torque - brake torque - tyre force times radius) divided by wheel inertia. It is retained exactly as supplied. The tyre-force sign in that wheel equation has not been independently confirmed against the missing Simulink model or a vehicle. Do not assume these plots establish real stopping-distance or traction-performance accuracy.

This revision changes presentation and explanations only. The original 1500 kg mass, 0.31 m radius, thresholds, fixed 1 ms step and torque-rate equations remain the reference defaults. Using a Genesis visual reference does not make those defaults a calibrated GV70 model.

Advanced equation reference

Signed slip equals (vehicle speed - tyre surface speed) divided by the maximum of their absolute speeds and 0.5 m/s. Brake slip is its positive part; drive slip is the positive part of its negative. The tyre curve uses s = minimum of absolute signed slip and 1.5, then clamps (1.28 times (1 - exp(-23.99 times s)) - 0.52 times s) divided by 1.10 between zero and one. Force equals minus the sign of signed slip times road friction times normal load times that curve.

ABS increases torque at 24,000 N m per second in APPLY and reduces it at 45,000 N m per second in RELEASE. TCS restores drive torque at 18,000 N m per second and reduces it at 35,000 N m per second. HOLD retains the previous applied value. Each applied torque is limited between zero and its current driver request. These are reference torque-rate rules, not valve or pump calibrations.

Road resistance equals 0.015 times mass times 9.81 plus 0.5 times 1.225 times 0.65 times speed squared, with speed in m/s. Euler integration advances speed, wheel rotation, distance and accumulated wheel angle by the fixed 0.001 s step. The implementation also clamps speed and wheel rotation rate to nonnegative values and stops both near standstill with sufficient brake demand.

Verify your experiments

Check A: run the default braking case. The mode sequence should contain 0, ABS apply, hold and release. Find a release sample: the diagram should show ABS RELEASE, the illustrative pump return path should be highlighted and the sensor panel should show the previous 1 ms feedback sample. Compare the selected applied brake torque with the torque graph.

Check B: run Braking without ABS. ABS active time should be zero; the pump teaching cue must remain off. A strong brake request may still create high brake slip. The absence of a pump animation does not mean the brake demand is zero.

Check C: run TCS low-friction launch. Find a TCS REDUCE sample. The drive command should be highlighted and drive torque must match the saved sample. The hydraulic pump must stay off. Run Launch without TCS with the same conditions and compare slip and torque. Do not compare runs with different road or speed settings unless that difference is your experiment.

Check D: change only playback speed or camera. Exported numerical results must remain identical. Change a physical input, such as road friction, and the old export buttons should disable until a new run completes. Set a step time beyond the duration: the interface should prevent running and explain the error.

For a lecturer exercise, give students the inputs and ask them to identify a release or reduction event, explain the feedback loop, quote its sample time and graph values, and state the limitations. CSV and JSON are local evidence files; they are not automatically submitted as an assessment.

Reference evidence and limits

Equations, thresholds and reference settings are from the supplied WheelSlip ABS/TCS implementation task and programmer guide, as documented in the project integration record. That record reports that the linked shared-chat model download failed and the source WheelSlip_ABS_TCS_Final.slx was unavailable for independent parity checks. This revision does not claim a new inspection of an unavailable source file.

This is a longitudinal, one-equivalent-wheel model with four equal tyre loads. It omits lateral motion, steering, yaw, load transfer, combined slip, real hydraulic actuators, communication delays and sensor failure dynamics. The pump cue and representative 3D hardware are explanatory layers. No ABS/TCS safety validation or exact Genesis control calibration is claimed.

Complete editable-input glossary

Simulation duration (s) | Default 12. How many seconds of driving the computer calculates. A 12-second run can be replayed faster or slower without changing the answer.

Brake step time (s) | Default 1. The simulated second when braking changes from the before value to the after value. For example, 1 means wait one simulated second, then change the brake demand.

Brake demand before step (ratio) | Default 0. Brake request before that change. Zero means no braking; 1 means the full maximum brake torque you entered.

Brake demand after step (ratio) | Default 1. Brake request from the brake step time onward. For example, 0.5 asks for half the maximum brake torque.

Throttle step time (s) | Default 1. The simulated second when the accelerator request changes from before to after. This is a scheduled change, not the calculation time step.

Throttle demand before step (ratio) | Default 0. Accelerator request before the scheduled change. Zero requests no drive torque; 1 requests the full maximum drive torque.

Throttle demand after step (ratio) | Default 0. Accelerator request after the scheduled change. Braking takes priority when brake demand exceeds 0.01.

Road friction coefficient (ratio) | Default 0.35. How strongly the tyre and road can grip. A smaller number represents a more slippery surface. This is a model setting, not a measurement of a particular road.

Initial vehicle speed (km/h) | Default 80. The starting speed of the vehicle and tyre surface. Both begin at the same speed, so initial slip is zero.

Vehicle mass (kg) | Default 1500. The total vehicle mass. The model divides its weight equally among four tyres; it does not move weight forward during braking.

Effective wheel radius (m) | Default 0.31. Distance from the wheel centre to the road. Radius times wheel rotation speed gives the tyre surface speed.

Wheel rotational inertia (kg·m²) | Default 1.2. How hard it is to change wheel rotation. A larger value makes the wheel resist speeding up or slowing down.

Maximum brake torque per wheel (N·m) | Default 1800. The largest braking twist requested at one wheel. The brake-demand fraction sets the request; ABS may change the applied amount.

Maximum drive torque per wheel (N·m) | Default 1200. The largest driving twist requested at one wheel. The throttle fraction sets the request; TCS may reduce the applied amount.

ABS enabled (on/off) | Default True. Allow the controller to increase, hold or release brake torque to manage braking slip. This model activates ABS only above 1 m/s vehicle speed.

TCS enabled (on/off) | Default True. Allow the controller to restore, hold or reduce drive torque when the driven wheel spins too quickly. This TCS model does not apply the hydraulic brakes.

Analytical signal definitions

Vehicle speed. How fast the model vehicle moves along the road. This is also the ideal vehicle-speed reference supplied to the controller.

Tyre surface speed. Wheel angular speed multiplied by radius, shown in km/h so it can be compared directly with vehicle speed.

Brake slip. How much slower the tyre surface moves than the vehicle, divided by the larger speed or 0.5 m/s near standstill. Zero means matched speeds; larger values show more braking slip.

Drive slip. How much faster the tyre surface moves than the vehicle, using the same normalisation. Larger values show more wheel spin.

Applied brake torque. The braking twist after the ABS decision and the requested-brake limit. It is not measured hydraulic pressure.

Applied drive torque. The driving twist after the TCS decision and the requested-drive limit.

Tyre force. Longitudinal force at one tyre contact. Negative is braking and positive is driving under the written model convention.

Controller mode. 0 means inactive; 1/2/3 mean ABS apply/hold/release; -1/-2/-3 mean TCS restore/hold/reduce. ABS has display priority if active.

ABS active. True includes apply, hold and release. It does not mean the return pump runs throughout every active sample.

TCS active. True includes restore, hold and reduce of drive torque; it does not imply hydraulic braking.

Distance. Total distance travelled from the start of this scenario.

Wheel angle. Integrated rotation used to position the 3D tyre, disc and encoder ring.

Distance travelled. Total distance at the end of the requested simulation duration.

Stopping time. First sample where vehicle speed is no more than 0.1 km/h; not reached means the run ended sooner.

Stopping distance. Travelled distance at the stopping-time sample.

Peak brake slip. Largest braking slip anywhere in the complete run.

Peak drive slip. Largest driving slip anywhere in the complete run.

Peak absolute tyre force. Largest force magnitude at the represented tyre, regardless of direction.

ABS active time. Added time in all ABS apply, hold and release modes.

TCS active time. Added time in all TCS restore, hold and reduce modes.

Final vehicle speed. Road speed at the last sample.

Final tyre surface speed. Wheel rotation converted to km/h at the last sample.